

NICHOLAS CARRERA

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UX designer and researcher with hands-on experience in usability evaluation, defect identification, and systems design. Proven ability to reframe ambiguous problems through structured research and deliver software that is functional, intuitive, and reliable.

EDUCATION

DePaul University — B.S. UX Design, Minor in Computer Science | GPA: 3.825

Chicago, IL | June 2026

PROJECTS

LLM-to-Geometry Research Pipeline DePaul URAP × Nathan Koch

Sep 2025 – Mar 2026

- Designed and ran prompt experiments across 4 LLMs (Mistral 7B, Claude Haiku/Sonnet/Opus), testing meta-cognitive prompting, noise variation, and constrained shape generation across 10 intensity levels
- Built a FastAPI server with multi-model provider abstraction, composite cache layer, and SSE streaming; integrated with TouchDesigner via WebSocket for a live browser → LLM → geometry render loop
- Designed and executed a controlled test matrix across 10 parameter levels, surfacing the 7-parameter prompt structure (not model reasoning) as the pipeline bottleneck; redirected research from model optimization to prompt design and approach expansion

Python · FastAPI · Anthropic Claude API · Ollama · TouchDesigner (Script SOP, WebSocket DAT) · SSE · WebSocket · LLM prompt engineering · procedural 3D geometry

CAT SSO & Access Control Redesign DePaul Capstone × Caterpillar

Jan 2026 – Jun 2026

- Conducted Nielsen heuristic evaluation of the SSO display and operator access controls across multiple Cat machine types, surfacing defects in terminology consistency (fast/standard/fine vs. coarse/normal/fine), iconography, and navigation hierarchy that broke operator workflows across job-sites
- Reframed the problem through on-site field research at Cat's Edwards facility, discovering that sub-5% Operator ID adoption was a trust deficit, not a UI failure, as operators defaulted to Guest Mode to avoid haste and reprimand; pivoted the team from UI optimization to a "designing backwards" approach centered on value exchange
- Designed a three-layer system architecture (identity, machine authorization, session) informed by Tesla and Toast SSO research, with a universal web portal, three sign-in methods (NFC tap, SMS code, 4-digit PIN) for redundancy, and a USB-sync fallback for offline connectivity

Figma · Nielsen heuristic evaluation · structured user-interviews · on-site field research · stakeholder interviews · systems architecture · userflow mapping · systems design

Urban Displacement Data Visualization Wabash Lights Public Installation

Mar 2025 – Jun 2025

- Designed a composite statistical metric (Renter-to-Owner Ratio Shift) from 6 UDP variables across Chicago census tracts, applying mean/median/std-dev analysis to produce an 8-level gentrification typology validated against ground-truth census data
- Built a P5.js animation mapping typology data to LED color behavior within the architectural constraints of the Wabash Lights, a public art canvas spanning the CTA elevated tracks in downtown Chicago
- Deployed live on the Wabash Lights installation, translating 28 years of housing displacement data into an impactful urban data visualization

P5.js · JavaScript · Tableau · statistical analysis · composite metric design · color theory · data ethics · public installation design

SKILLS

Development: Python, JavaScript, TypeScript, React, FastAPI, P5.js, WebSocket, SSE, Node.js, Java, Scala

Design & Research: Figma, heuristic evaluation, structured user-interviews, affinity mapping, persona development, systems mapping, journey mapping, data visualization, usability testing · bug reporting · test case design · accessibility evaluation

AI / ML: Anthropic Claude API, Ollama, LLM prompt engineering, multi-model evaluation, procedural 3D geometry